The Transformer encoder, by hand

Worksheet #66464 · sentence “red the big” · n = 3, d_model = 4, h = 2, d_k = 2, d_ff = 8, one encoder layer · from Attention by Hand

Name _____ Date _____

Write one number in each empty box, rounded to 2 decimals, and use your rounded answers in the steps after. Shaded tables are given to you. A basic calculator helps for √, sin, cos and e^x (use radians). Every step has a hint at the bottom, and the answer key is on the last pages.

Step 1: Token IDs

Look up each word of **“red the big”** in the vocabulary.

Vocabulary: id, word, embedding row

	d0	d1	d2	d3
1 big	-1	0	1	-1
2 ran	0	-1	1	0
3 red	0	0	-1	0
4 sat	-1	-1	-1	0
5 the	-1	-1	0	-1

IDs, shape (3,)

red	the	big

Hint. Find each word in the vocabulary. The number in front of it is its ID. No arithmetic yet.

Step 2: Embedding lookup E

`E = the vocabulary row of each ID · shape 3 × 4`

E (3 × 4)

	d0	d1	d2	d3
red				
the				
big				

Hint. Copy the whole row of each word's ID. Row “red” of E is the vocabulary row of “red”.

Step 3: Scale by √d_model

`E′ = E × √4 = E × 2`

E′ (3 × 4)

	d0	d1	d2	d3
red				
the				
big				

Hint. Every entry gets the same multiplier. E′[red, d0] = E[red, d0] × 2.

Step 4: Positional encoding PE

$PE[pos, 2i] = \sin(pos \div 10000^{2i/d}) \cdot PE[pos, 2i+1] = \cos(\text{same angle})$

divisor for each pair

pair 0	pair 1

PE (3 × 4)

	d0 sin	d1 cos	d2 sin	d3 cos
pos 0 red				
pos 1 the				
pos 2 big				

Hint. Divisor for pair i = $10000^{2i/4}$. Pair 0: $10000^{0} = 1$. Pair 1: $10000^{0.5} = \sqrt{10000}$. Then angle = pos ÷ divisor; take sin for the even column and cos for the odd one, in radians. The pos 0 row is always sin 0 = 0 and cos 0 = 1.

Step 5: Add position to meaning: X = E′ + PE

$X = E' + PE \cdot \text{shape } 3 \times 4$

X (3 × 4)

	d0	d1	d2	d3
red				
the				
big				

Hint. Add the numbers in the same row and column: X[red, d1] = E′[red, d1] + PE[pos 0, d1].

Step 6: Head 1: queries, keys and values

$Q = X \cdot W_Q,\ K = X \cdot W_K,\ V = X \cdot W_V \cdot (3 \times 4) \cdot (4 \times 2) = 3 \times 2$

W_Q

	0	1
d0	0	0
d1	0	0
d2	0	1
d3	0	0

W_K

	0	1
d0	0	0
d1	0	0
d2	1	1
d3	1	1

W_V

	0	1
d0	-1	-1
d1	-1	0
d2	-1	1
d3	-1	1

Q

	0	1
red		
the		
big		

K

	0	1
red		
the		
big		

V

	0	1
red		
the		
big		

Hint. Each entry is one row of X dotted with one column of the weights: Q[red, 0] = X[red, d0]×0 + X[red, d1]×0 + X[red, d2]×0 + X[red, d3]×0.

Step 7: Head 1: scores $S = Q \cdot K^T$

`S[r → c] = Q[r] · K[c] · shape 3 × 3`

S (rows: query token, columns: key token)

	red	the	big
red			
the			
big			

Hint. S[red → the] = Q[red, 0]×K[the, 0] + Q[red, 1]×K[the, 1]. Row r says how well token r's query matches every key.

Step 8: Head 1: scale by √d_k

`S ÷ √2 = S ÷ 1.41`

S ÷ 1.41

	red	the	big
red			
the			
big			

Hint. Divide every score by 1.41. This keeps the softmax in the next step from getting too extreme.

Step 9: Head 1: softmax, row by row

`A[r, c] = e^x[r, c] ÷ (sum of e^x across row r)`

e^x of each scaled score

	red	the	big
red			
the			
big			

row total

red	
the	
big	

A

	red	the	big
red			
the			
big			

Hint. Take e^x of every entry (the e^x key on a calculator). For reference: $e^0 = 1$, $e^{0.5} \approx 1.65$, $e^1 \approx 2.72$, $e^{-1} \approx 0.37$. Add each row to get its total, then divide each entry by its row total. Check: every row of A adds up to 1.

Step 10: Head 1: blend the values, A · V

`head output = A · V · (3 × 3) · (3 × 2) = 3 × 2`

Head 1 output

	0	1
red		
the		
big		

Hint. out[red, 0] = A[red → red]×V[red, 0] + A[red → the]×V[the, 0] + A[red → big]×V[big, 0]. Each row is a weighted average of the value rows.

Step 11: Head 2: queries, keys and values

`Q = X · W_Q, K = X · W_K, V = X · W_V · (3 × 4) · (4 × 2) = 3 × 2`

W_Q

	0	1
d0	0	0
d1	0	0
d2	0	-1
d3	0	0

W_K

	0	1
d0	1	-1
d1	1	0
d2	0	0
d3	0	0

W_V

	0	1
d0	-1	-1
d1	0	-1
d2	0	1
d3	-1	1

Q

	0	1
red		
the		
big		

K

	0	1
red		
the		
big		

V

	0	1
red		
the		
big		

> **Hint.** Each entry is one row of X dotted with one column of the weights: Q[red, 0] = X[red, d0]×0 + X[red, d1]×0 + X[red, d2]×0 + X[red, d3]×0.

Step 12: Head 2: scores S = Q · K^T

`S[r → c] = Q[r] · K[c] · shape 3 × 3`

S (rows: query token, columns: key token)

	red	the	big
red			
the			
big			

> **Hint.** S[red → the] = Q[red, 0]×K[the, 0] + Q[red, 1]×K[the, 1]. Row r says how well token r's query matches every key.

Step 13: Head 2: scale by √d_k

`S ÷ √2 = S ÷ 1.41`

S ÷ 1.41

	red	the	big
red			
the			
big			

> **Hint.** Divide every score by 1.41. This keeps the softmax in the next step from getting too extreme.

Step 14: Head 2: softmax, row by row

`A[r, c] = e^x[r, c] ÷ (sum of e^x across row r)`

e^x of each scaled score

	red	the	big
red			
the			
big			

row total

red	
the	
big	

A

	red	the	big
red			
the			
big			

Hint. Take e^x of every entry (the e^x key on a calculator). For reference: $e^0 = 1$, $e^{0.5} \approx 1.65$, $e^1 \approx 2.72$, $e^{-1} \approx 0.37$. Add each row to get its total, then divide each entry by its row total. Check: every row of A adds up to 1.

Step 15: Head 2: blend the values, A · V

`head output = A · V · (3 × 3) · (3 × 2) = 3 × 2`

Head 2 output

	0	1
red		
the		
big		

Hint. out[red, 0] = A[red → red]×V[red, 0] + A[red → the]×V[the, 0] + A[red → big]×V[big, 0]. Each row is a weighted average of the value rows.

Step 16: Combine the heads with W_O

`MHA = Concat(heads) · W_O · (3 × 4) · (4 × 4) = 3 × 4`

Concat

	h1·0	h1·1	h2·0	h2·1
red				
the				
big				

W_O

	d0	d1	d2	d3
d0	1	1	0	0
d1	0	-1	0	0
d2	1	0	0	-1
d3	1	0	0	0

MHA

	d0	d1	d2	d3
red				
the				
big				

Hint. Concat: place the head outputs side by side, head 1 first. Then MHA[red, d0] = Concat[red, 0]×1 + Concat[red, 1]×0 + Concat[red, 2]×1 + Concat[red, 3]×1.

Step 17: Add & Norm (1)

`R1 = X + MHA, then X1 = (R1 - row mean) ÷ row sd`

R1 = X + MHA

	d0	d1	d2	d3
red				
the				
big				

mean

red	
the	
big	

sd

red	
the	
big	

X1

	d0	d1	d2	d3
red				
the				
big				

Hint. Row mean = (sum of the 4 numbers) ÷ 4. Row sd = √(average of (value − mean)²), dividing by 4. Then X1[r, c] = (R1[r, c] − mean of row r) ÷ sd of row r. γ = 1 and β = 0 here, so nothing else changes. Check: each row of X1 adds up to about 0.

Step 18: Feed-forward: expand, then ReLU

`P = X1 · W1 + b1 · (3 × 4) · (4 × 8) = 3 × 8, then H = max(0, P)`

W1

	h0	h1	h2	h3	h4	h5	h6	h7
d0	0	-1	-1	0	0	1	0	-1
d1	0	0	1	1	1	0	0	0
d2	-1	0	0	1	0	0	0	0
d3	0	1	-1	0	0	0	-1	0

b1

h0	h1	h2	h3	h4	h5	h6	h7
0	0	0	0	0	1	0	0

P = X1 · W1 + b1

	h0	h1	h2	h3	h4	h5	h6	h7
red								
the								
big								

H = ReLU(P)

	h0	h1	h2	h3	h4	h5	h6	h7
red								
the								
big								

Hint. P[red, h0] = X1[red, d0]×0 + X1[red, d1]×0 + X1[red, d2]×(−1) + X1[red, d3]×0 + 0. ReLU keeps positive numbers and writes 0 for negative ones.

Step 19: Feed-forward: compress back

`F = H · W2 + b2 · (3 × 8) · (8 × 4) = 3 × 4`

W2

	d0	d1	d2	d3
h0	1	0	0	1
h1	0	1	1	0
h2	0	0	-1	1
h3	1	0	1	1
h4	0	0	0	0
h5	0	1	0	0
h6	0	0	0	0
h7	0	0	0	-1

b2

d0	d1	d2	d3
0	0	0	0

F

	d0	d1	d2	d3
red				
the				
big				

Hint. F[red, d0] = H[red, h0]×1 + H[red, h1]×0 + H[red, h2]×0 + H[red, h3]×1 + H[red, h4]×0 + H[red, h5]×0 + H[red, h6]×0 + H[red, h7]×0 + 0. Every 0 in H wipes out its term, so skip those.

Step 20: Add & Norm (2): the encoder output Z

`R2 = X1 + F, then Z = (R2 - row mean) ÷ row sd`

R2 = X1 + F

	d0	d1	d2	d3
red				
the				
big				

mean

red	
the	
big	

sd

red	
the	
big	

Z (3 × 4)

	d0	d1	d2	d3
red				
the				
big				

Hint. The same recipe as Add & Norm (1), with X1 as the residual.

You pushed "red the big" through a whole encoder layer by hand. One row per word went in, and one row per word came out, each now mixed with context from the others.

Answer key · worksheet #66464

Worked with every value rounded to 2 decimals, the same way you did. If you only differ in the last digit, that is rounding.

IDs

red	3
the	5
big	1

E

	d0	d1	d2	d3
red	0	0	-1	0
the	-1	-1	0	-1
big	-1	0	1	-1

E′

	d0	d1	d2	d3
red	0	0	-2	0
the	-2	-2	0	-2
big	-2	0	2	-2

divisors

pair 0	pair 1
1	100

PE

	d0	d1	d2	d3
pos 0 red	0	1	0	1
pos 1 the	0.84	0.54	0.01	1
pos 2 big	0.91	-0.42	0.02	1

X

	d0	d1	d2	d3
red	0	1	-2	1
the	-1.16	-1.46	0.01	-1
big	-1.09	-0.42	2.02	-1

head 1 Q

	0	1
red	0	-2
the	0	0.01
big	0	2.02

head 1 K

	0	1
red	-1	-1
the	-0.99	-0.99
big	1.02	1.02

head 1 V

	0	1
red	0	-1
the	3.61	0.17
big	0.49	2.11

head 1 S

	red	the	big
red	2	1.98	-2.04
the	-0.01	-0.01	0.01
big	-2.02	-2	2.06

head 1 S ÷ 1.41

	red	the	big
red	1.42	1.4	-1.45
the	-0.01	-0.01	0.01
big	-1.43	-1.42	1.46

head 1 e^x

	red	the	big
red	4.14	4.06	0.23
the	0.99	0.99	1.01
big	0.24	0.24	4.31

total

red	8.43
the	2.99
big	4.79

head 1 A

	red	the	big
red	0.49	0.48	0.03
the	0.33	0.33	0.34
big	0.05	0.05	0.9

head 1 output

	0	1
red	1.75	-0.35
the	1.36	0.44
big	0.62	1.86

head 2 Q

	0	1
red	0	2
the	0	-0.01
big	0	-2.02

head 2 K

	0	1
red	1	0
the	-2.62	1.16
big	-1.51	1.09

head 2 V

	0	1
red	-1	-2
the	2.16	1.63
big	2.09	2.53

head 2 S

	red	the	big
red	0	2.32	2.18
the	0	-0.01	-0.01
big	0	-2.34	-2.2

head 2 S ÷ 1.41

	red	the	big
red	0	1.65	1.55
the	0	-0.01	-0.01
big	0	-1.66	-1.56

head 2 e^x

	red	the	big
red	1	5.21	4.71
the	1	0.99	0.99
big	1	0.19	0.21

total

red	10.92
the	2.98
big	1.4

head 2 A

	red	the	big
red	0.09	0.48	0.43
the	0.34	0.33	0.33
big	0.71	0.14	0.15

head 2 output

	0	1
red	1.85	1.69
the	1.06	0.69
big	-0.09	-0.81

Concat

	h1·0	h1·1	h2·0	h2·1
red	1.75	-0.35	1.85	1.69
the	1.36	0.44	1.06	0.69
big	0.62	1.86	-0.09	-0.81

MHA

	d0	d1	d2	d3
red	5.29	2.1	0	-1.85
the	3.11	0.92	0	-1.06
big	-0.28	-1.24	0	0.09

R1

	d0	d1	d2	d3
red	5.29	3.1	-2	-0.85
the	1.95	-0.54	0.01	-2.06
big	-1.37	-1.66	2.02	-0.91

mean

red	1.39
the	-0.16
big	-0.48

sd

red	2.94
the	1.43
big	1.47

X1

	d0	d1	d2	d3
red	1.33	0.58	-1.15	-0.76
the	1.48	-0.27	0.12	-1.33
big	-0.61	-0.8	1.7	-0.29

P

	h0	h1	h2	h3	h4	h5	h6	h7
red	1.15	-2.09	0.01	-0.57	0.58	2.33	0.76	-1.33
the	-0.12	-2.81	-0.42	-0.15	-0.27	2.48	1.33	-1.48
big	-1.7	0.32	0.1	0.9	-0.8	0.39	0.29	0.61

H

	h0	h1	h2	h3	h4	h5	h6	h7
red	1.15	0	0.01	0	0.58	2.33	0.76	0
the	0	0	0	0	0	2.48	1.33	0
big	0	0.32	0.1	0.9	0	0.39	0.29	0.61

F

	d0	d1	d2	d3
red	1.15	2.33	-0.01	1.16
the	0	2.48	0	0
big	0.9	0.71	1.12	0.39

R2

	d0	d1	d2	d3
red	2.48	2.91	-1.16	0.4
the	1.48	2.21	0.12	-1.33
big	0.29	-0.09	2.82	0.1

mean

red	1.16
the	0.62
big	0.78

sd

red	1.64
the	1.35
big	1.19

Z

	d0	d1	d2	d3
red	0.8	1.07	-1.41	-0.46
the	0.64	1.18	-0.37	-1.44
big	-0.41	-0.73	1.71	-0.57